Quantum Mechanics Homework 5 Solutions

Lectures 10–12

Useful formulas

$$\int_{-\infty}^{\infty} e^{-bx^2}\,dx = \sqrt{\frac{\pi}{b}}, \tag{1}$$

$$\int_{-\infty}^{\infty} x^2 e^{-bx^2}\,dx = \frac{\sqrt{\pi}}{2b^{3/2}}, \tag{2}$$

$$\int_{-\infty}^{\infty} x e^{-bx^2}\,dx = 0. \tag{3}$$

Problem 1. An even state in a finite square well [20 points]

(a) At $x = a/2$, continuity of the wave function gives

$$A\cos\left(\frac{ka}{2}\right) = De^{-\beta(a/2-a/2)} = D. \tag{4}$$

Therefore

$$\boxed{D = A\cos\left(\frac{ka}{2}\right).} \tag{5}$$

(b) The derivatives on the two sides of the boundary are

$$\phi'_{\text{in}}(a/2) = -Ak\sin\left(\frac{ka}{2}\right), \tag{6}$$

$$\phi'_{\text{out}}(a/2) = -\beta D. \tag{7}$$

Continuity of ϕ' gives

$$-Ak\sin\left(\frac{ka}{2}\right) = -\beta D. \tag{8}$$

Dividing this equation by $A\cos(ka/2) = D$ gives

$$-k\tan\left(\frac{ka}{2}\right) = -\beta, \tag{9}$$

so

$$\boxed{k\tan\left(\frac{ka}{2}\right) = \beta.} \tag{10}$$

(c) If $ka/2 = \pi/4$, then

$$\tan\left(\frac{ka}{2}\right) = \tan\left(\frac{\pi}{4}\right) = 1. \tag{11}$$

The matching condition therefore gives

$$\boxed{\beta = k.} \tag{12}$$

From the definitions,

$$\frac{E}{V_0} = \frac{k^2}{k^2 + \beta^2}. \tag{13}$$

Since $\beta = k$,

$$\boxed{\frac{E}{V_0} = \frac{k^2}{2k^2} = \frac{1}{2}.} \tag{14}$$

(d) From $ka/2 = \pi/4$,

$$k = \frac{\pi}{2a}. \tag{15}$$

Since $\beta = k$,

$$\boxed{\ell = \frac{1}{\beta} = \frac{2a}{\pi}.} \tag{16}$$

The state is even because its interior solution is $\cos(kx)$. In the interval $-a/2 < x < a/2$, the argument satisfies $-\pi/4 < kx < \pi/4$, where the cosine has no zero. Thus

$$\boxed{\text{even parity, zero interior nodes.}} \tag{17}$$

Problem 2. Power-series termination for the harmonic oscillator [20 points]

The termination condition for a polynomial of degree n is

$$2n + 1 - \lambda = 0, \qquad \lambda_n = 2n + 1. \tag{18}$$

(a) For $n = 0$,

$$\lambda_0 = 1, \qquad E_0 = \frac{\lambda_0}{2}\hbar\omega = \boxed{\frac{1}{2}\hbar\omega}. \tag{19}$$

The polynomial is a constant:

$$H_0(\xi) = a_0. \tag{20}$$

Hence $\psi_0 \propto e^{-\xi^2/2}$, which is even and has no node:

$$\boxed{\psi_0 : \text{ even, zero nodes.}} \tag{21}$$

(b) For $n = 1$,

$$\lambda_1 = 3, \qquad E_1 = \frac{3}{2}\hbar\omega. \tag{22}$$

The polynomial is

$$H_1(\xi) = a_1\xi. \tag{23}$$

Thus

$$\boxed{\psi_1 : \text{ odd, one node at } \xi = 0.} \tag{24}$$

(c) For $n = 2$,

$$\lambda_2 = 5, \qquad E_2 = \boxed{\frac{5}{2}\hbar\omega}. \tag{25}$$

Setting $j = 0$ in the recurrence relation gives

$$a_2 = \frac{1 - \lambda_2}{2} a_0 = -2a_0. \tag{26}$$

Therefore

$$H_2(\xi) = a_0(1 - 2\xi^2) \propto 2\xi^2 - 1. \tag{27}$$

The Gaussian factor is never zero, so the nodes are determined by the polynomial:

$$2\xi^2 - 1 = 0 \quad \Longrightarrow \quad \boxed{\xi = \pm\frac{1}{\sqrt{2}}}. \tag{28}$$

(d) Using $E_2 = 5\hbar\omega/2$,

$$\frac{1}{2}m\omega^2 x_\pm^2 = \frac{5}{2}\hbar\omega, \tag{29}$$

$$x_\pm^2 = \frac{5\hbar}{m\omega}. \tag{30}$$

Hence

$$\boxed{x_\pm = \pm\sqrt{\frac{5\hbar}{m\omega}}}. \tag{31}$$

The wave function is mainly oscillatory between these turning points and has decaying tails outside them.

Problem 3. Gaussian ground state and expectation values [20 points]

(a) Normalization requires

$$1 = \int_{-\infty}^{\infty} |\psi_0(x)|^2 \, dx \tag{32}$$

$$= A^2 \int_{-\infty}^{\infty} e^{-\alpha^2 x^2} \, dx \tag{33}$$

$$= A^2 \frac{\sqrt{\pi}}{\alpha}. \tag{34}$$

Therefore

$$A^2 = \frac{\alpha}{\sqrt{\pi}}, \qquad \boxed{A = \frac{\sqrt{\alpha}}{\pi^{1/4}}.} \tag{35}$$

(b) The expectation value is

$$\langle x \rangle = A^2 \int_{-\infty}^{\infty} x e^{-\alpha^2 x^2} \, dx. \tag{36}$$

The probability density is even and x is odd, so the complete integrand is odd. Its integral over symmetric limits vanishes:

$$\boxed{\langle x \rangle = 0.} \tag{37}$$

(c) Using the supplied integral with $b = \alpha^2$,

$$\langle x^2 \rangle = A^2 \int_{-\infty}^{\infty} x^2 e^{-\alpha^2 x^2} \, dx \tag{38}$$

$$= \frac{\alpha}{\sqrt{\pi}} \frac{\sqrt{\pi}}{2\alpha^3} \tag{39}$$

$$= \frac{1}{2\alpha^2}. \tag{40}$$

Since $\alpha^2 = m\omega/\hbar$,

$$\boxed{\langle x^2 \rangle = \frac{\hbar}{2m\omega}.} \tag{41}$$

Because $\langle x \rangle = 0$,

$$\boxed{\Delta x = \sqrt{\langle x^2 \rangle} = \sqrt{\frac{\hbar}{2m\omega}}.} \tag{42}$$

(d) The potential-energy expectation value is

$$\langle V \rangle = \frac{1}{2} m\omega^2 \langle x^2 \rangle \tag{43}$$

$$= \frac{1}{2} m\omega^2 \left(\frac{\hbar}{2m\omega} \right) \tag{44}$$

$$= \boxed{\frac{1}{4}\hbar\omega}. \tag{45}$$

Since $E_0 = \langle T \rangle + \langle V \rangle = \hbar\omega/2$,

$$\langle T \rangle = \frac{1}{2}\hbar\omega - \frac{1}{4}\hbar\omega = \boxed{\frac{1}{4}\hbar\omega}. \tag{46}$$

Thus the ground-state energy is shared equally between average kinetic and potential energy.

Problem 4. Factorization and adjoints [20 points]

(a) The product of the two scales is

$$x_0 p_0 = \sqrt{\frac{2\hbar}{m\omega}}\sqrt{2\hbar m\omega} = \boxed{2\hbar}. \tag{47}$$

Also,

$$\hbar\omega\left(\frac{\hat{p}}{p_0}\right)^2 = \hbar\omega\frac{\hat{p}^2}{2\hbar m\omega} = \frac{\hat{p}^2}{2m}, \tag{48}$$

$$\hbar\omega\left(\frac{\hat{x}}{x_0}\right)^2 = \hbar\omega\frac{m\omega\hat{x}^2}{2\hbar} = \frac{1}{2}m\omega^2\hat{x}^2. \tag{49}$$

Therefore

$$\boxed{\hat{H} = \hbar\omega\left[\left(\frac{\hat{p}}{p_0}\right)^2 + \left(\frac{\hat{x}}{x_0}\right)^2\right].} \tag{50}$$

(b) Expanding in the original order gives

$$\hat{a}^\dagger\hat{a} = \left(\frac{\hat{x}}{x_0} - i\frac{\hat{p}}{p_0}\right)\left(\frac{\hat{x}}{x_0} + i\frac{\hat{p}}{p_0}\right) \tag{51}$$

$$= \frac{\hat{x}^2}{x_0^2} + \frac{\hat{p}^2}{p_0^2} + \frac{i}{x_0 p_0}(\hat{x}\hat{p} - \hat{p}\hat{x}) \tag{52}$$

$$= \frac{\hat{x}^2}{x_0^2} + \frac{\hat{p}^2}{p_0^2} + \frac{i}{x_0 p_0}[\hat{x}, \hat{p}]. \tag{53}$$

Using $[\hat{x}, \hat{p}] = i\hbar$ and $x_0 p_0 = 2\hbar$,

$$\frac{i}{x_0 p_0}[\hat{x}, \hat{p}] = \frac{i(i\hbar)}{2\hbar} = -\frac{1}{2}. \tag{54}$$

Hence

$$\hat{a}^\dagger\hat{a} = \frac{\hat{x}^2}{x_0^2} + \frac{\hat{p}^2}{p_0^2} - \frac{1}{2}, \tag{55}$$

and

$$\boxed{\hat{H} = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right).} \tag{56}$$

The additional $1/2$ comes from the nonzero position–momentum commutator.

(c) Since x is real,

$$\int f^*(\hat{x}^\dagger g)\,dx = \int (\hat{x}f)^* g\,dx \tag{57}$$

$$= \int (xf)^* g\,dx \tag{58}$$

$$= \int f^*(xg)\,dx \tag{59}$$

$$= \int f^*(\hat{x}g)\,dx. \tag{60}$$

Since this holds for all functions in the domain,

$$\boxed{\hat{x}^\dagger = \hat{x}.} \tag{61}$$

(d) Beginning with the adjoint definition,

$$\int f^* \left[\left(\frac{d}{dx}\right)^\dagger g\right] dx = \int \left(\frac{df}{dx}\right)^* g\,dx \tag{62}$$

$$= \int \frac{df^*}{dx} g\,dx \tag{63}$$

$$= [f^*g]_{-\infty}^{\infty} - \int f^* \frac{dg}{dx}\,dx. \tag{64}$$

Since f and g vanish at both ends,

$$[f^*g]_{-\infty}^{\infty} = 0. \tag{65}$$

Therefore

$$\boxed{\left(\frac{d}{dx}\right)^\dagger = -\frac{d}{dx}.} \tag{66}$$

The boundary condition is needed to remove the boundary term. Without it, the formal integration-by-parts result would contain an additional contribution, and the operator domain would have to be reconsidered.